Calculating aggregates

Lesson 2.4.3

Foundational methods for deriving mobility and network insights from Mobile Phone Data

Contents

- Dimensions of aggregates (selecting temporal and spatial resolutions)
- The common types of aggregate and their variations:
    - Presence
    - Trips and travellers
    - Residents and relocations
    - Social mixing
    - Network status and usage
- Calculating selected short-term (e.g. daily) aggregates
- Calculating selected long-term (e.g. monthly) aggregates
- Calculating common network status and usage aggregates
- Combining aggregates

What are aggregates?

Aggregates

In previous sessions, we have computed statistics about *individual* (pseudonymised) subscribers:

e.g.

- continuity models
- meaningful locations
- usual environments
- home locations

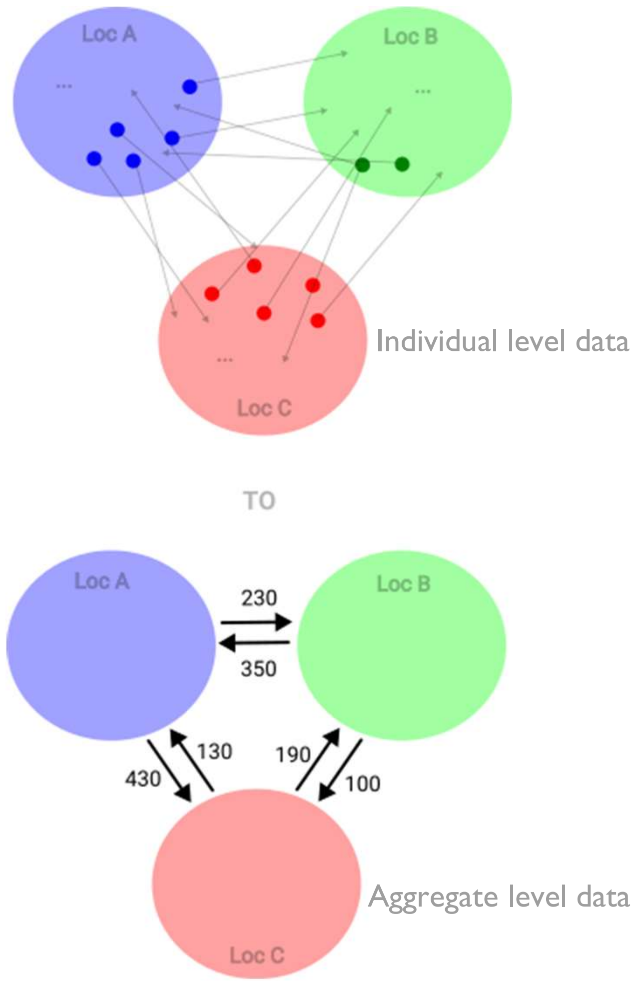
1 → 2 → 3 → 4

1. Individual level analysis (e.g. meaningful locations)
2. Creating aggregates
3. Aggregates to Estimates
4. Actionable insights

We need to combine to reveal *group* statistics / patterns of behaviour (also, for privacy reasons we shouldn't reveal statistics that can identify an individual person)

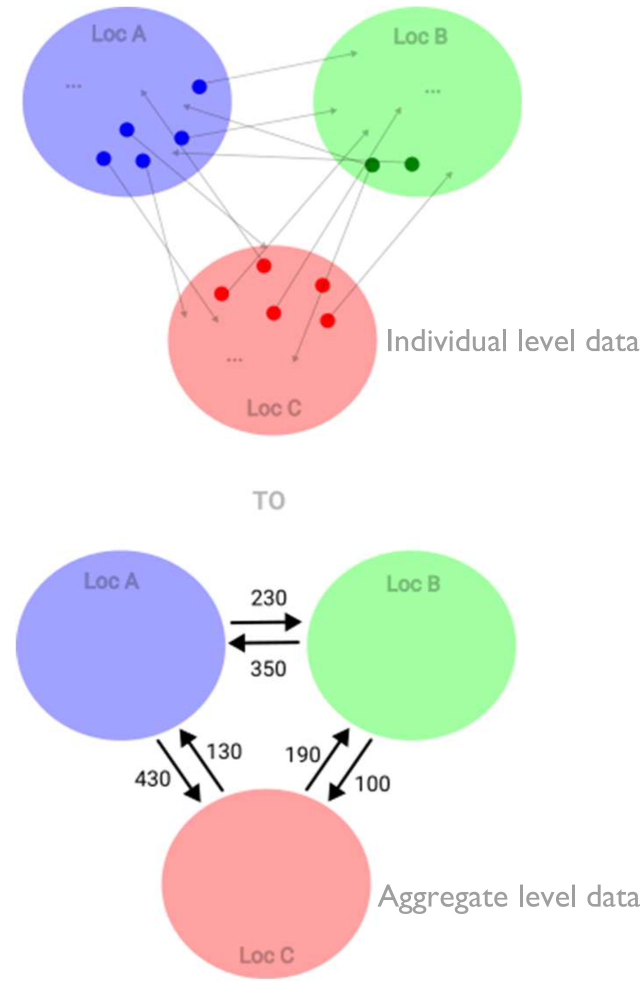
Objectives of an aggregate

A mobility aggregate turns the **pseudonymised CDR data of many individual subscribers** into an output that characterises the **behaviour of the entire group of subscribers without revealing information about specific individuals**.



Objectives of an aggregate

- Fully anonymous
  - All aggregates should be privacy compliant and not allow the identification of individual subscribers
- Easy to compute (especially in crisis scenarios)
  - Depending on the model / context, the person who wrote the query may not be the one executing it (e.g. if using an assisted queries approach and the MNO is producing the aggregates)
- Robust to infrequent phone usage
  - Aggregates produced to understand mobility should be driven by mobility (and not phone usage)



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Dimensions of aggregates

Dimensions of aggregates

How CDR aggregates are broken down in space (spatial resolution) and time (temporal resolution) is key to which types of questions they can inform.

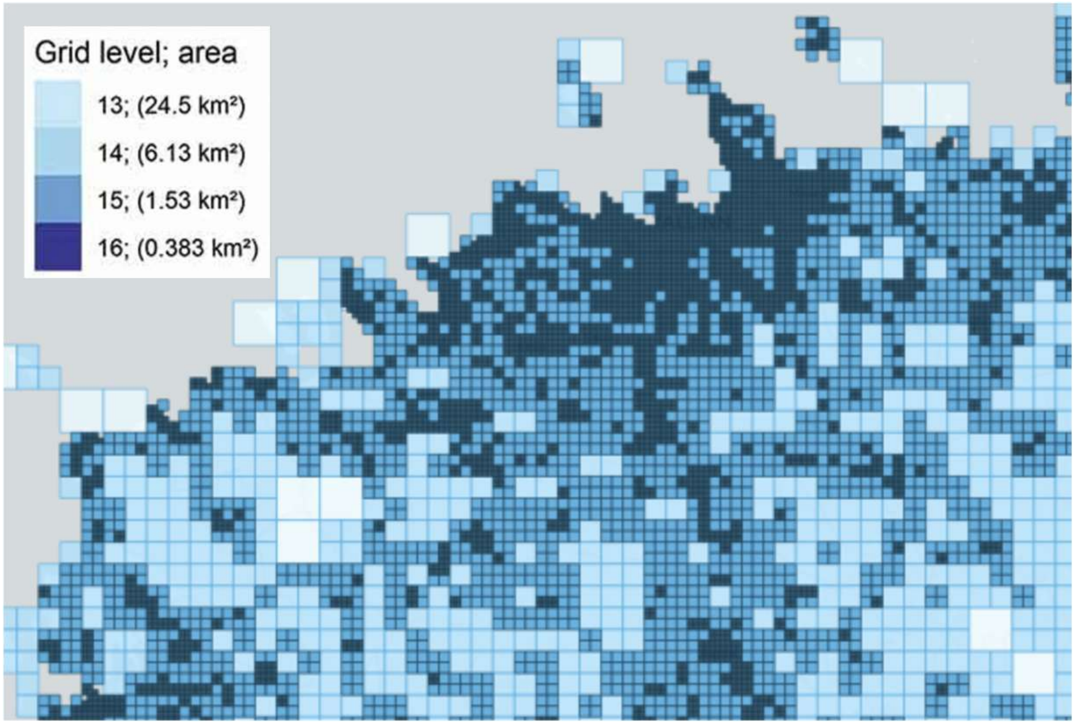
…The questions being asked usually dictate which dimensions we should use when calculating aggregates!

Spatial resolution

- Which resolution you create aggregates at depends on the questions being asked

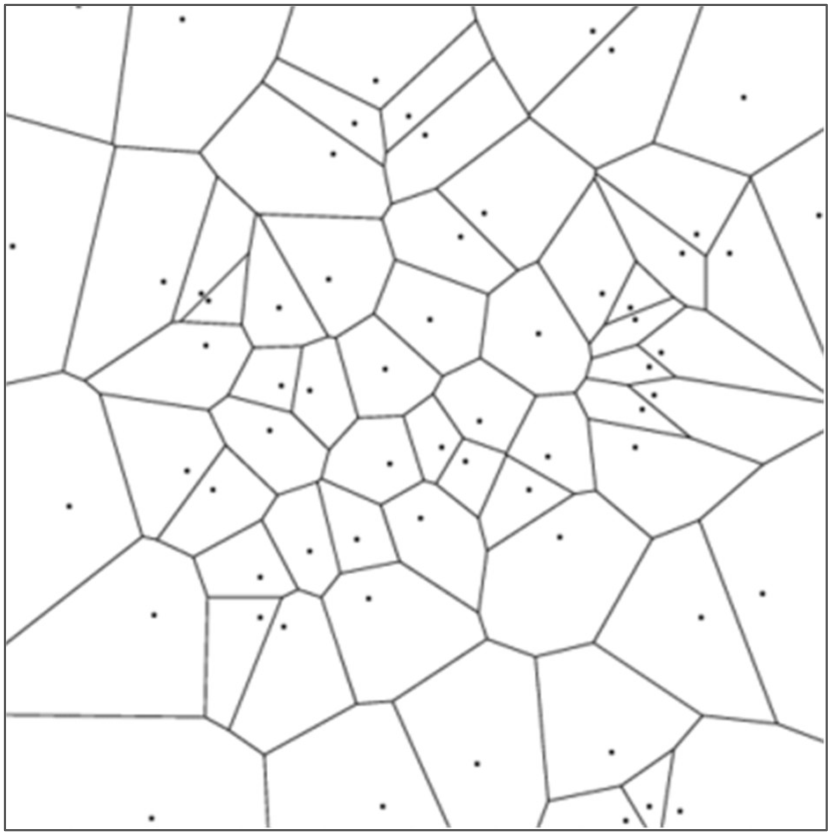
e.g. do you need to work at a high spatial resolution if you want to compare MPD derived statistics to other statistics calculated at a lower granularity?

Aggregation. Spatial: Grid

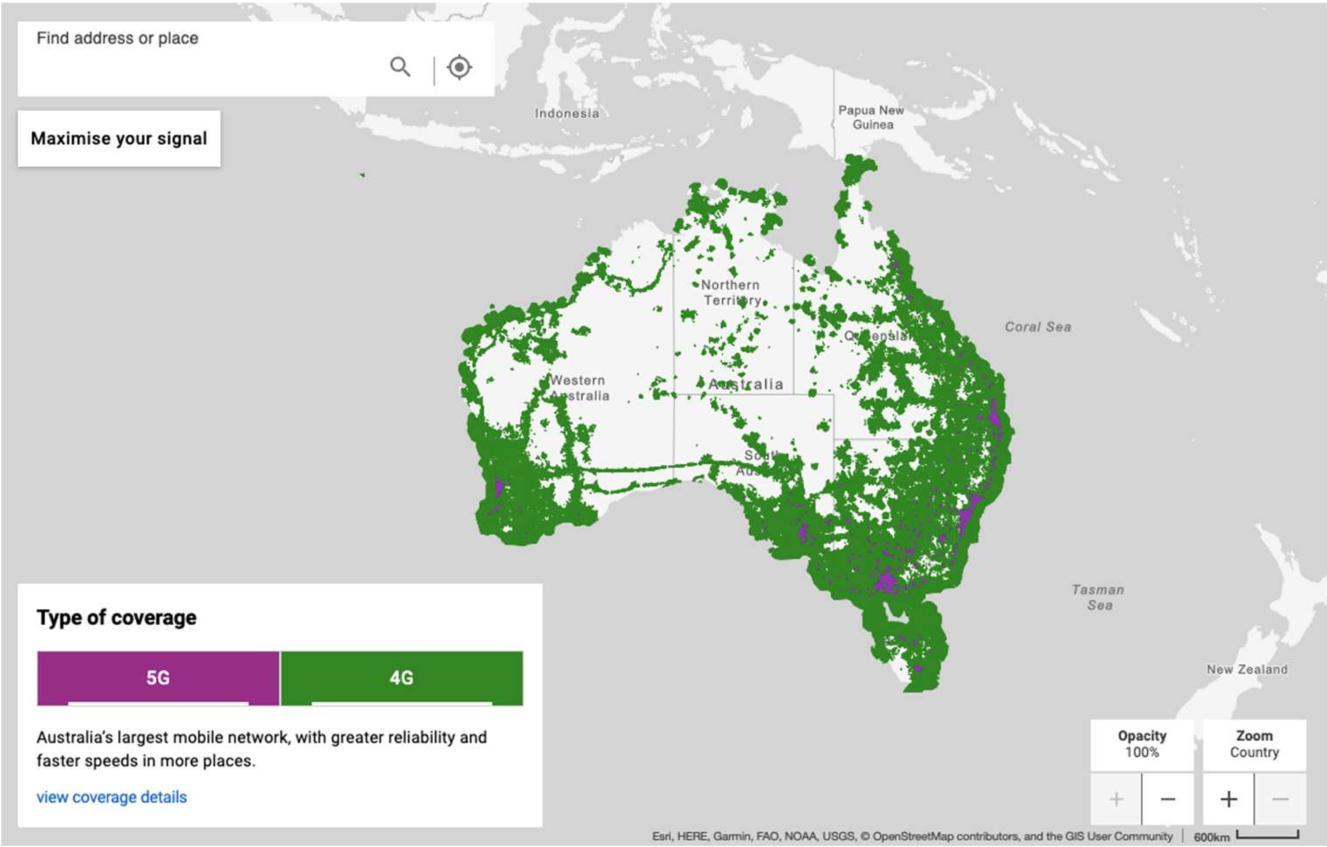


Adaptive grid example from Tallinn Metropolitan Area from a research paper "Spatial interpolation of mobile positioning data for population statistics" https://doi.org/10.1080/17489725.2021.1917710

Aggregation. Spatial: Voronois

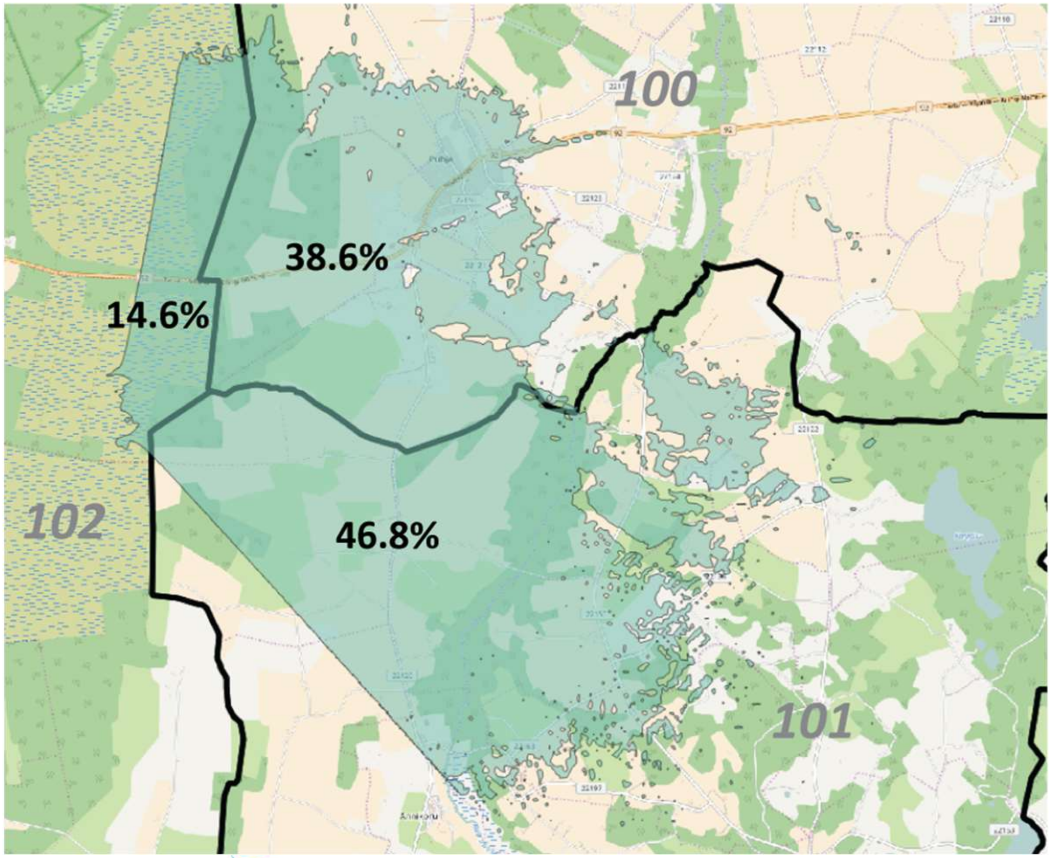


Aggregation. Spatial: Cell coverage maps



https://www.telstra.com.au/coverage-networks/our-coverage

Aggregation. Spatial: Local Administrative Unit (LAU)



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Note: the more granular we go (in both spatial and temporal resolution), the more likely we will need to redact data due to being under the minimum reporting threshold for privacy

150	430
110	200

(890)

30	10	40	80
70	50	190	120
5	90	50	50
5	10	90	10

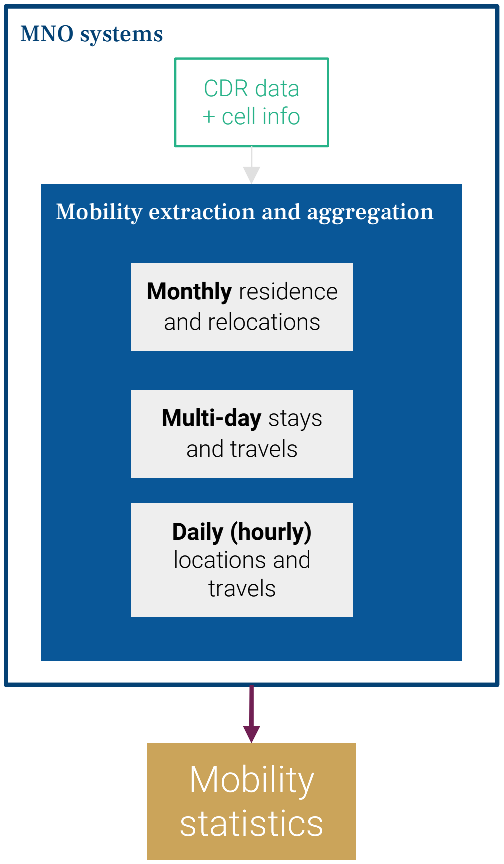
(860)

Temporal resolution

- Which resolution you create aggregates at depends on the questions being asked

e.g. Do you need high temporal resolution aggregates to compute changes in usual subscriber residence? Is monthly or yearly enough to answer your questions?

Categories of indicators per time scale



Time scale and type	Comparability and possibility for adjustments
Residence location: Dominant unique location per month	**Easier to compare to existing data**, census and survey data, for adjustments and validation
Stay location: Dominant unique location per X days (e.g. per week)	Fewer existing data to compare it to
Visited locations: Number of subscribers visiting a location in a day (subscribers can visit more than one location in a day)	Few survey data exists on this type of mobility - possibility to compare with app data

Forming aggregates

Forming aggregates

1. Compute individual level statistics (previous sessions)
2. Decide which geometry to use for the specific use case (can be done at the same time)
   a. Link geometry to features computed per subscriber at the chosen geometry level
3. Aggregate by geometry (usually counting), might want to take other statistics such as medians / 25th + 75th percentiles
4. Redact / add privacy preserving measures if appropriate

Forming aggregates (example)

Individual level statistics

msisdn	estimated_location	timestamp
FE61520BCA21E52	cluster_01	26/11/2025
FE61520BCA21E52	cluster_03	27/11/2025
FE61520BCA21E52	cluster_04	28/11/2025
…	…	…

in this particular instance we want to compute aggregates at the administrative level

Forming aggregates (example)

Cell cluster to (less granular geometry) mapping

cell_cluster	adm3_pcode	adm3_en
cluster_01	mc000001	Bluetown
cluster_03	mc000001	Redtown
cluster_04	mc002011	Bluetown
…	…	…

Note that two different administrative regions have the same name (this can happen, more likely at the more granular administrative level). Always use a unique identifier when joining spatial information, such as the pcode.

Forming aggregates (example)

Individual level statistics with cluster -> pcode mapping

msisdn	estimated_location	timestamp
FE61520BCA21E52	cluster_01	26/11/2025
FE61520BCA21E52	cluster_03	27/11/2025
FE61520BCA21E52	cluster_04	28/11/2025
…	…	…

adm3_pcode
mc000001
mc000001
mc002011
…

Forming aggregates (example)

Individual level statistics with cluster -> pcode mapping

msisdn	estimated_location	timestamp	adm3_pcode
FE61520BCA21E52	cluster_01	26/11/2025	mc000001
FE61520BCA21E52	cluster_03	27/11/2025	mc000001
FE61520BCA21E52	cluster_04	28/11/2025	mc002011
…	…	…	…

For each timestamp, count the occurrences of the attached spatial information to form the aggregate

Forming aggregates (example)

Redact below 15

Aggregates

timestamp	adm3_pcode	count
~~26/11/2025~~	~~mc000001~~	~~10~~
27/11/2025	mc000001	20
28/11/2025	mc000001	105
…	…	…

Information about specific subscribers now removed.

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Doing this in practise

- Depends entirely on the data access agreement you have with the MNO:

Direct access to the CDR data?

- SQL / Python / PySpark / …

Aggregates are produced by someone at the MNO?

- SQL often easiest

Common aggregates

Commonly used aggregates

- Presence
- Trips and travellers
- Social mixing
- Residents and relocations
- Network status and usage

Short term aggregates

Presence

<u>Time dimension</u>

Hourly / Daily aggregate.

<u>Spatial dimension:</u>

Cell level / cluster level / administrative level

- Can be defined in multiple ways:
  - How many people were present at Location X during time period T? ← *More general, but lacks precision, can double count people if they appeared at multiple locations in the provided time period*
  - How many people are currently estimated to be located at location X at time Y? ← *Needs a good continuity model, and good phone usage / coverage in the area*

Presence (daily)

```sql
-- Simple presence aggregate: unique subscribers per location per day

SELECT call_date, location_id, COUNT(DISTINCT msisdn) AS subscriber_count
FROM calls
WHERE call_date BETWEEN :start_date::DATE AND :end_date::DATE
GROUP BY call_date, location_id
HAVING COUNT(DISTINCT msisdn) >= 15; -- Privacy threshold
```

Note: In a day, if someone makes calls at different times at different locations, they will be ‘present’ in all of those locations

Presence (hourly, no double counting)

```sql
-- Presence aggregate with continuity model
-- For each hour, use the most recent location for each subscriber
WITH most_recent_locations AS
(
	SELECT
		DATE_TRUNC('hour', call_datetime) AS hour_start,
		msisdn, location_id,
		call_datetime,
		ROW_NUMBER() OVER ( PARTITION BY DATE_TRUNC('hour', call_datetime), msisdn ORDER BY call_datetime DESC ) AS rn
	FROM calls
	WHERE call_datetime BETWEEN :start_datetime::TIMESTAMP AND :end_datetime::TIMESTAMP -- Label each subscribers calls per hour
),
hourly_presence AS
(
	SELECT hour_start, msisdn, location_id
	FROM most_recent_locations WHERE rn = 1 -- Keep only the most recent location per subscriber per hour
)
SELECT hour_start, location_id, COUNT(DISTINCT msisdn) AS subscriber_count
FROM hourly_presence
GROUP BY hour_start, location_id
HAVING COUNT(DISTINCT msisdn) >= 15; -- Privacy threshold
```

Presence (hourly, no double counting)

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```

Trips and Travellers

<u>Temporal dimension:</u>
Hourly / Daily

<u>Spatial dimension:</u>
Cell, Cell cluster, Administrative unit

Method:

1) Compute diary of stops per subscriber (see Continuity models section)
    a) A diary of stops may look like [Loc A, Loc B, Loc A, Loc A, Loc B]
2) Count the consecutive stops per subscriber
    a) [Loc A, Loc B], [Loc B, Loc A], [Loc A, Loc A], [Loc A, Loc B]
3) Aggregate flows
    a) [Loc A, Loc A] : 1, [Loc A, Loc B] : 2, [Loc B, Loc A] : 1

- Caveat: may not capture long distance trips if subscribers stop while travelling.

Trips

```
WITH consecutive_stops AS
(
        SELECT
                msisdn,
                stop_datetime,
                location_id,
                LEAD(location_id) OVER ( PARTITION BY msisdn ORDER BY stop_datetime ) AS next_location_id,
                LEAD(stop_datetime) OVER ( PARTITION BY msisdn ORDER BY stop_datetime ) AS next_stop_datetime
        FROM stops
        WHERE stop_datetime BETWEEN :start_datetime::TIMESTAMP AND :end_datetime::TIMESTAMP
),

…
```

Per subscriber join subsequent stops

Trips

```
…

trips AS (
    SELECT
        DATE_TRUNC('day', stop_datetime) AS trip_date,
        msisdn,
        location_id AS origin_location,
        next_location_id AS destination_location
    FROM consecutive_stops
    WHERE next_location_id IS NOT NULL -- Exclude last stop (no next location)
    AND location_id != next_location_id -- Only count actual moves )

…
```

In this instance we are counting trips per day, only need the locations and the datetime of when they left (or arrived, depending on the context)

Trips

```
…

SELECT
	trip_date,
	origin_location,
	destination_location,
	COUNT(DISTINCT msisdn) AS trip_count
FROM trips
GROUP BY trip_date, origin_location, destination_location
HAVING COUNT(DISTINCT msisdn) >= 15; -- Privacy threshold
```

Count the number of trips made per corridor, per day

Is one spatial / temporal dimension enough?

Depending on the questions being asked, maybe not!

Consider the question ‘Are the same people visiting Location X throughout the week or is there a large turnover of subscribers?’

This logic was used to find areas where there might be a higher risk of COVID-19 spreading between people from different communities

Consider comparing the **number unique subscribers present at any point in a given day**
to the **number of unique subscribers present at any point in a given week.** This tells us if there an area is visited by different groups of subscribers or the same people consistently - which can be used as an indicator for which areas are more likely to spread diseases.

Long term aggregates

Home locations

Discussed in previous session. One computed per subscriber, aggregated it counts the number of assigned home locations in a given window.

Time dimension:

Weekly, Monthly, Greater than Monthly

Spatial dimension:

Cell cluster level / Administrative level

- Can be computed in several ways depending on criteria for home location
- For crisis applications more granular → Can see changes in home location faster

Poll

After running a home detection algorithm, Location A has 1000 subscriber home locations in Month 1, and then 2000 subscriber home locations in Month 2. What could this mean?

A) 1000 people moved to Location A from other Locations on Month 2
B) Some people left location A but more people arrived at location A and the total net change equals +1000 people
C) Something else
D) All of the above

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Poll

After running a home detection algorithm, Location A has 1,000 subscriber home locations in Month 1, and then 2,000 subscriber home locations in Month 2. What could this mean?

A) 1,000 people moved to Location A from other Locations on Month 2 ← *could happen*
B) Some people left location A but more people arrived at location A and the total net change equals +1,000 people ← *could happen*
C) Something else← *more often the case*
D) All of the above

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Only counting subscribers at a location is not always useful.

(instead we count changes in location and add them over time)

Changes in home location counts between two consecutive months = Changes in mobility + Changes in phone usage

Number of subscribers at Location A at Month 2

+ Number of people who moved to Location A from elsewhere from Month 1 → Month 2 →

Number of Subscribers who stayed in consecutive months

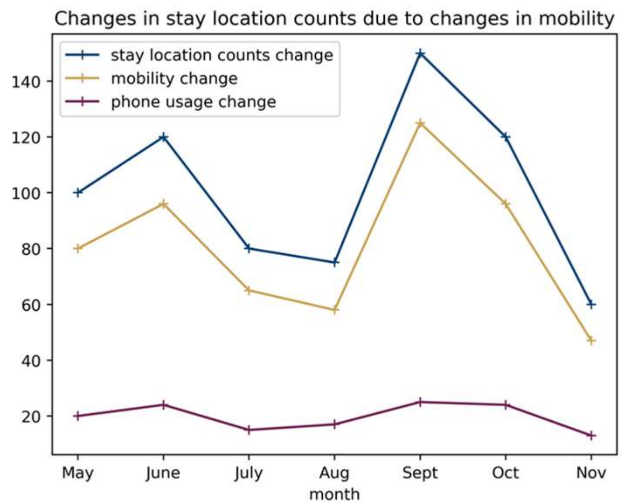
+ Number of subscribers who ‘appeared’ (driven by usage)
- Number of subscribers who ‘disappeared’ (driven phone usage)

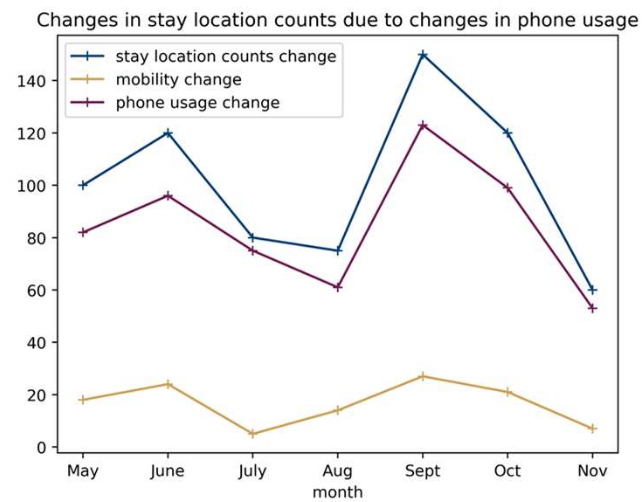
→ - Number of people who moved from Location A elsewhere from Month 1 → Month 2

Only counting subscribers at a location is not always useful.

(instead we count changes in location and add them over time)

Changes in stay location counts between two consecutive months	=	Changes in mobility	+	Changes in phone usage





Illustrative examples

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Only counting subscribers at a location is not always useful.

(instead we count changes in location and add them over time)

Changes in stay location counts between two consecutive months	=	Changes in mobility	+	Changes in phone usage

Subscribers at Location A at Month 2

\+ Number of people who moved to Location A from elsewhere from Month 1 → Month 2 →

Number of Subscribers who stayed in consecutive months

\+ ~~Number of subscribers who 'appeared'~~

\- ~~Number of subscribers who 'disappeared'~~

→ - Number of people who moved from Location A elsewhere from Month 1 → Month 2

Depending on context, continuity model, and phone usage, these factors can be very noisy

Using aggregates to make new aggregates

As a result Flowminder computes monthly population statistics via:

```
Month 1 pop: Baseline pop + (scaled) total inflows – (scaled) total outflows

Month 2 pop: Month 1 + (scaled) total inflows – (scaled) total outflows

…

Month X pop: Month (X-1) + (scaled) total inflows – (scaled) total outflows
```

Now the change is driven by movements we observe rather than people dropping in and out of the network

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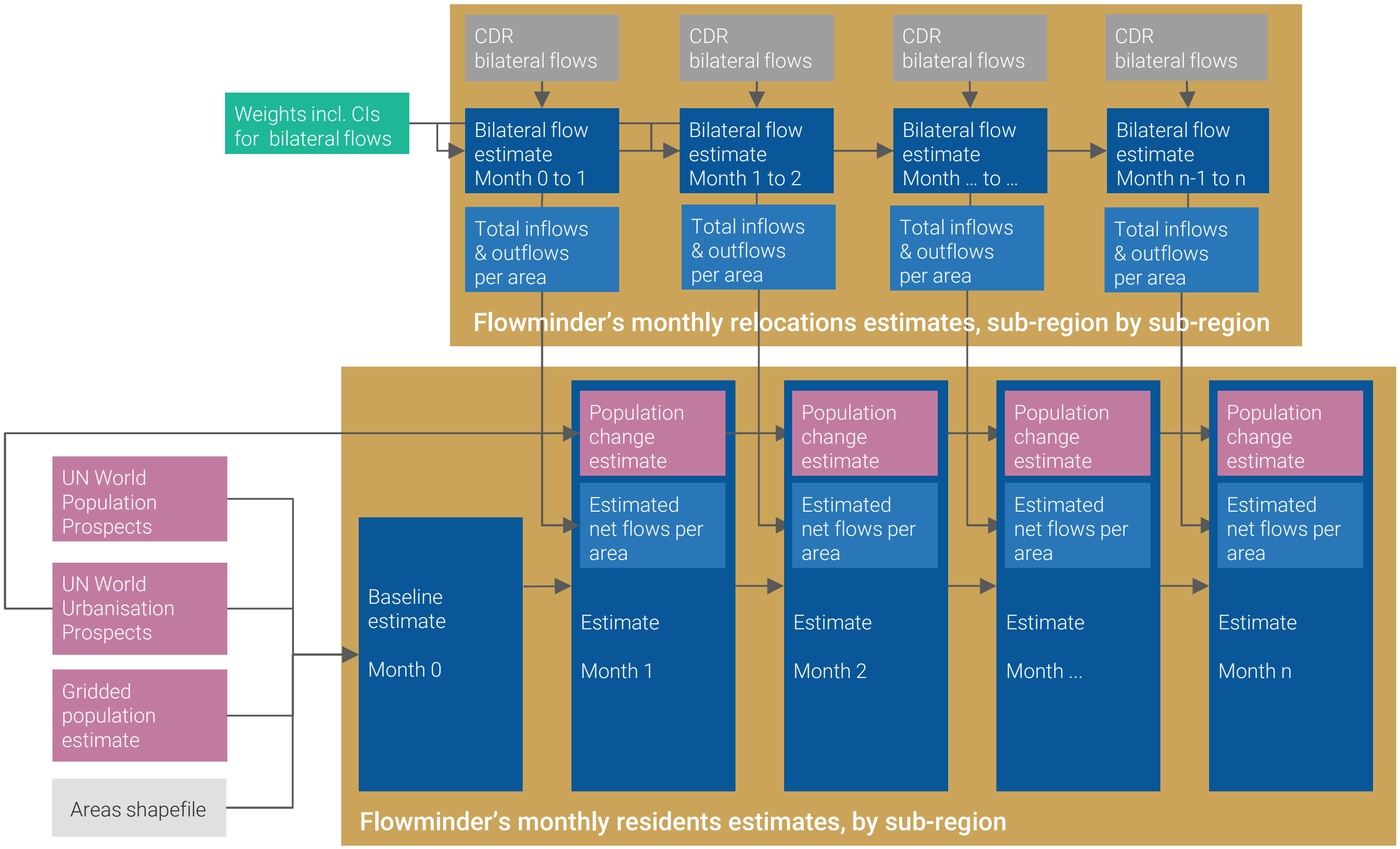
A method for monthly population estimates



Network / usage aggregates

Network usage aggregates

When producing aggregates to answer specific questions, we need to be careful about how we interpret them.

- Auxiliary aggregates that determine whether what we see is driven by mobility, or changes in the phone network.

- **For example:** changes in home locations may not be driven by actual mobility, but instead network outages.

- **For example:** Some areas may appear to have more frequent trips between them compared to a different region, but this could be driven by people making more CDR events in one area over the other.

Active cells

<u>Temporal scale:</u>

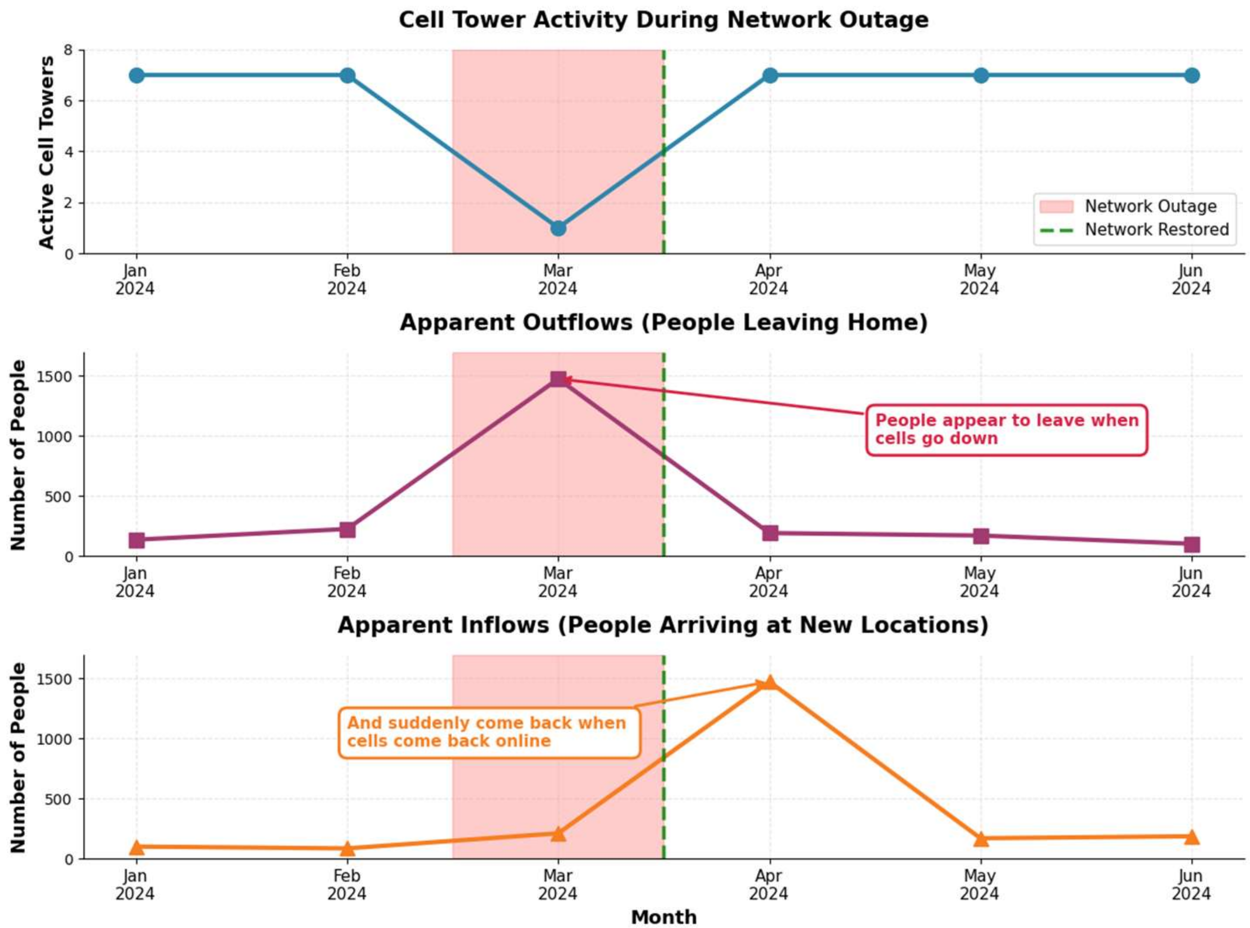
Daily

<u>Spatial scale:</u>

Cell cluster level / Administrative level

Method:

- Count the number of unique locations attached to each CDR record each temporal unit, per spatial unit



Cell Tower Activity During Network Outage
Active Cell Towers
Network Outage
Network Restored
Apparent Outflows (People Leaving Home)
Number of People
People appear to leave when cells go down
Apparent Inflows (People Arriving at New Locations)
And suddenly come back when cells come back online
Jan 2024
Feb 2024
Mar 2024
Apr 2024
May 2024
Jun 2024
Month

Total events

Temporal scale:

Daily / hourly

Spatial scale:

Cell cluster level / Administrative level

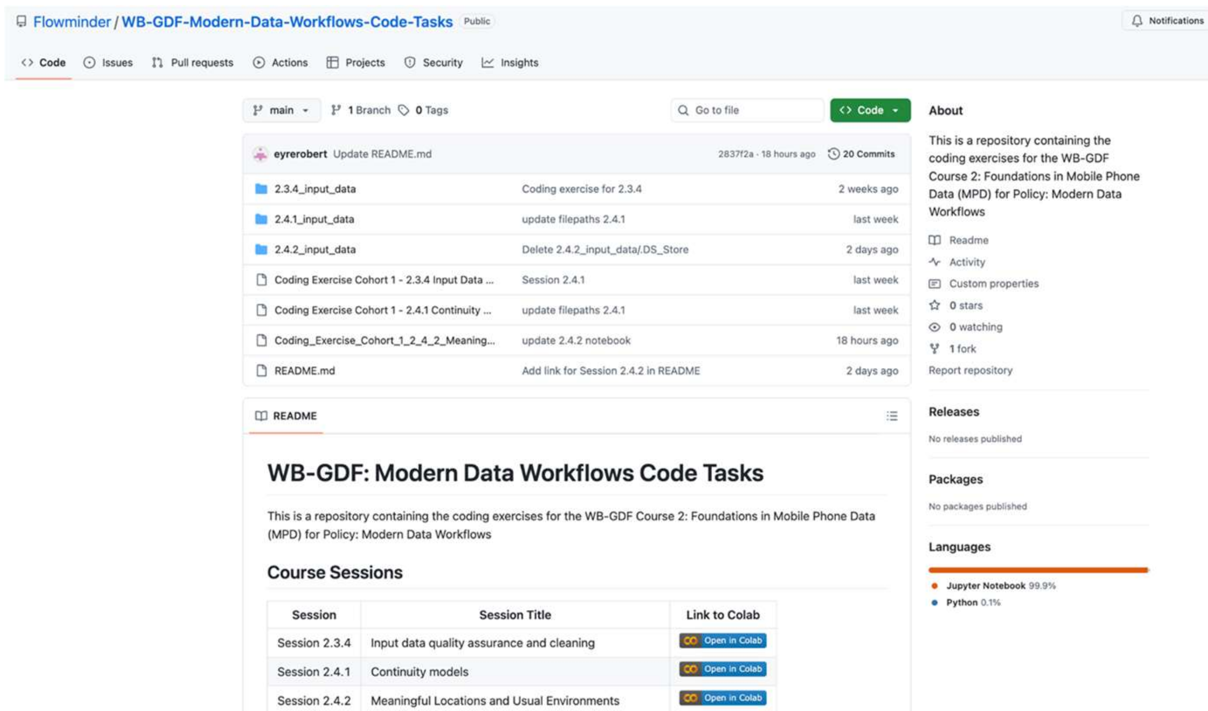
Method:

- Count the number of events per temporal unit, per spatial unit

Introduction to Practical Exercise

Calculating aggregates: Intro to Practical Exercise

- **Task**: Compute basic aggregates.
- **Exercise Link**: https://github.com/Flowminder/WB-GDF-Modern-Data-Workflows-Code-Tasks



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